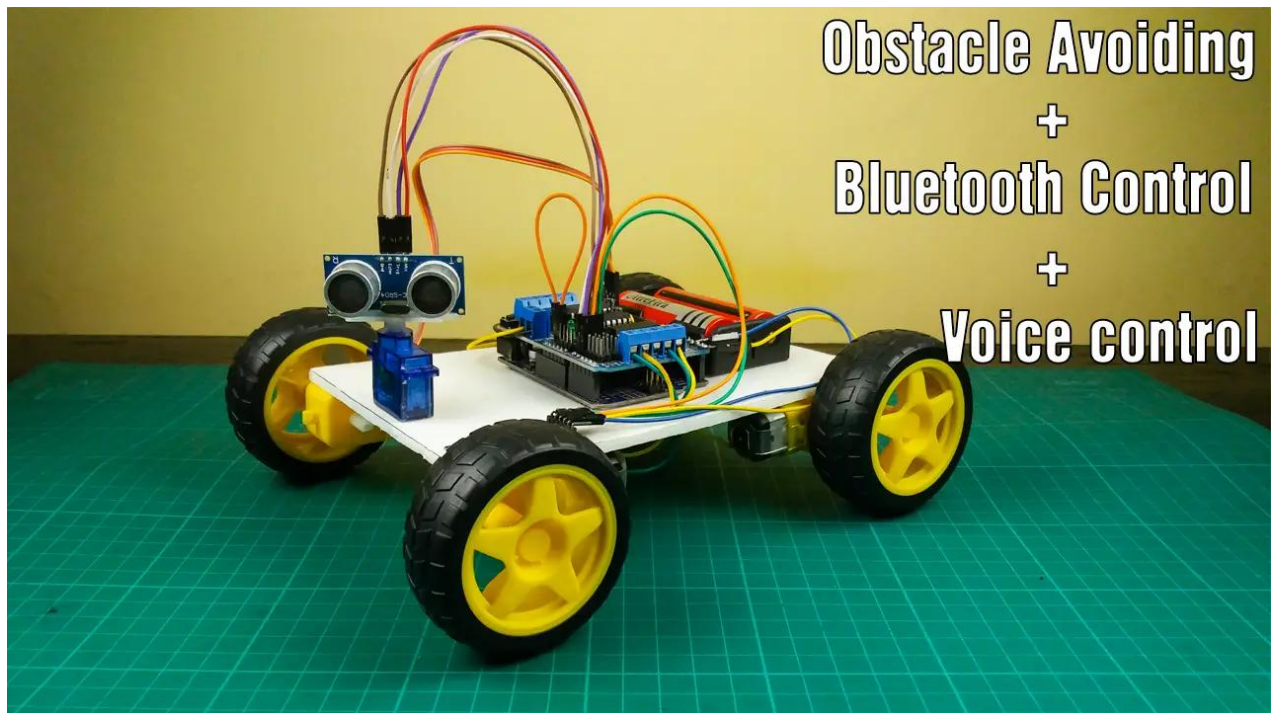


Multi-functional Arduino robot



The process of this robot

We can control this robot car using three methods. That is,

1.Obstacle avoidance

In this case, the robot car moves along using the [obstacle avoiding](#). The ultrasonic sensor is mainly used for this purpose. Study the previous articles for more information.

2.Bluetooth control

In this case, we can control the robot through an app on the smartphone. The Bluetooth module is used for this.

3.Voice control

In this case, we can control this robot using several voice commands. This also requires a Bluetooth module and mobile app.

OK,let's do this project step by step. The required components are given below.

Step 1

Firstly, identify these components.



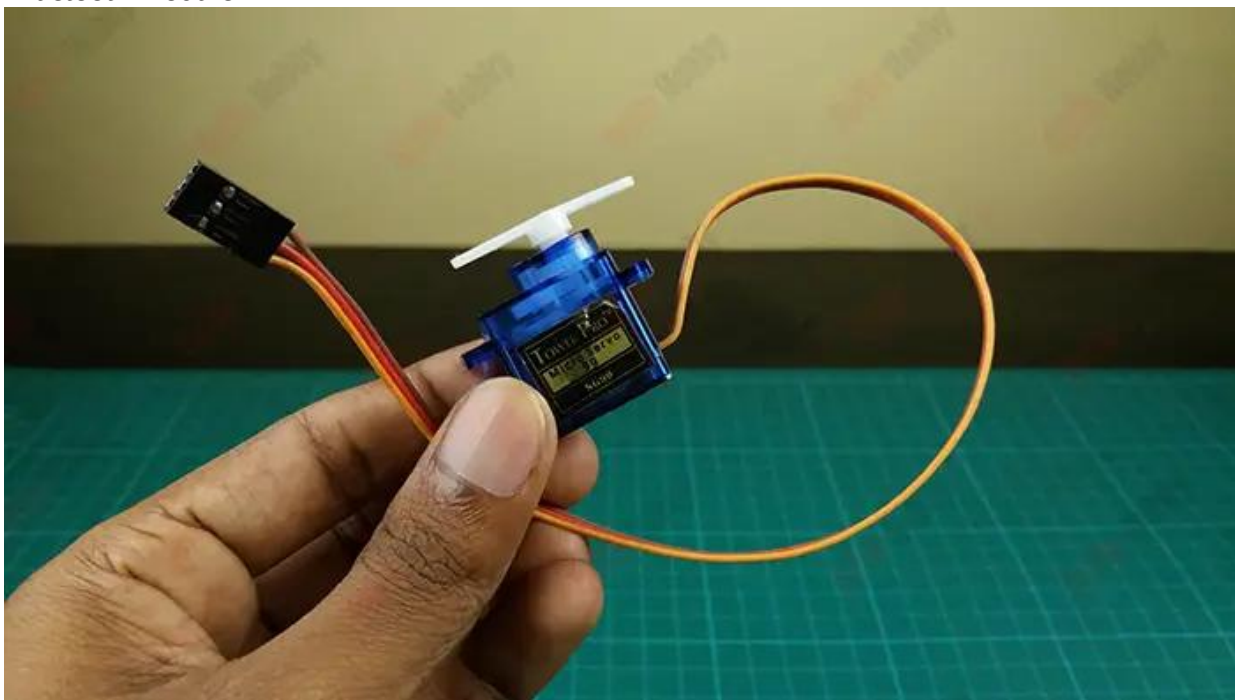
L293D motor driver shield



Ultrasonic sensor



Bluetooth module



Servo motor



Li-ion battery



Li-ion battery



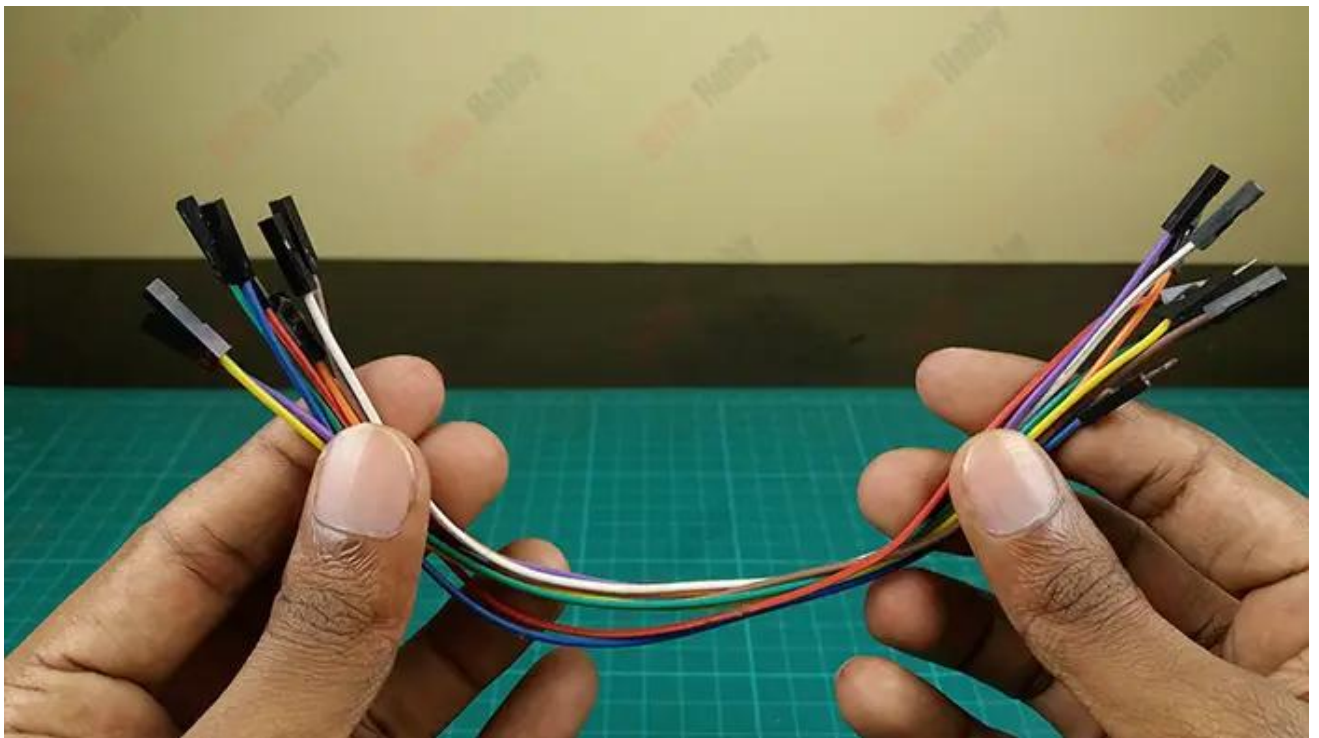
Foamboard



Robot wheel



Gear motor



Jumper wires



Arduino UNO board

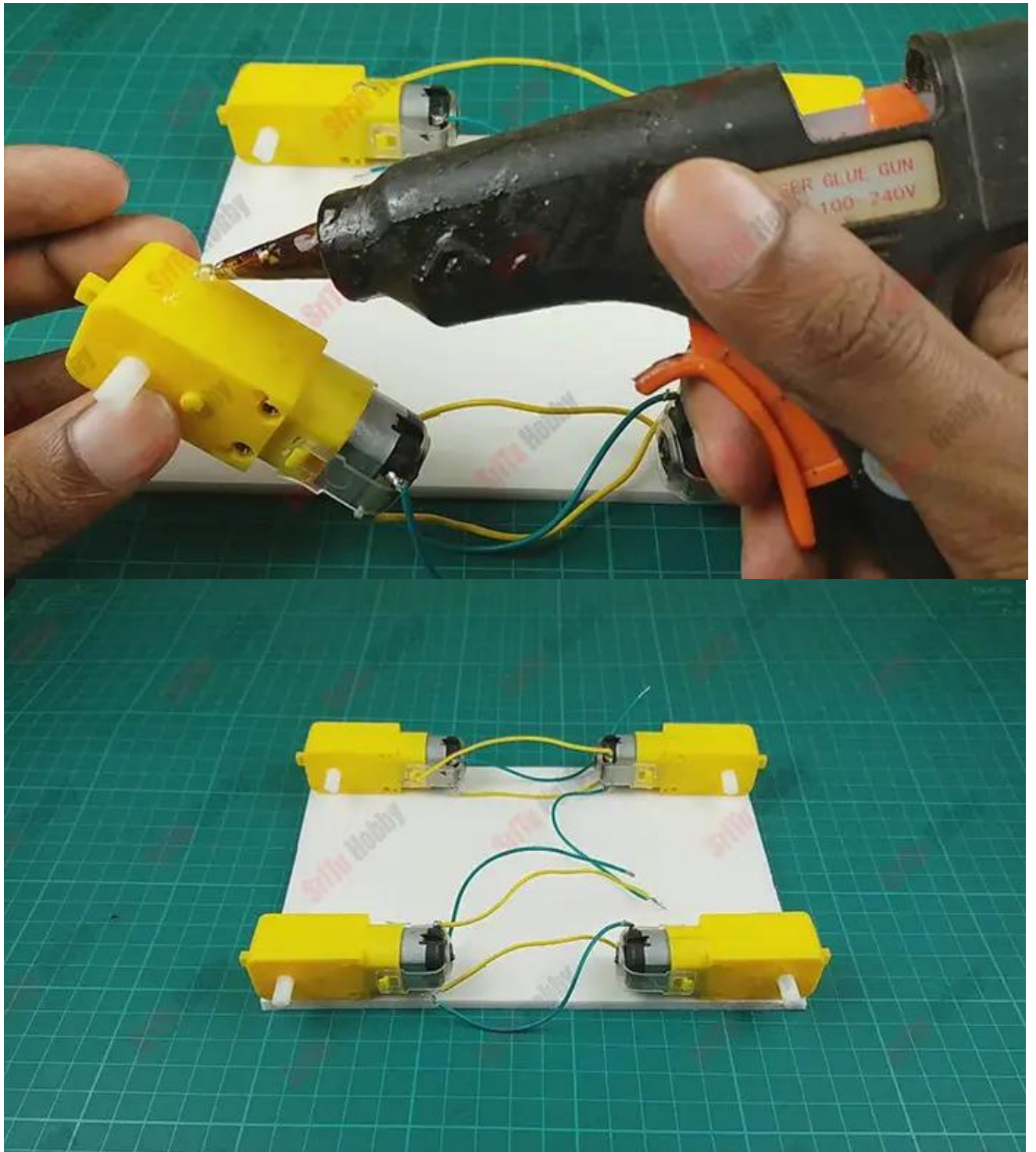
Step 2

Secondly, cut the foam board piece as follows.



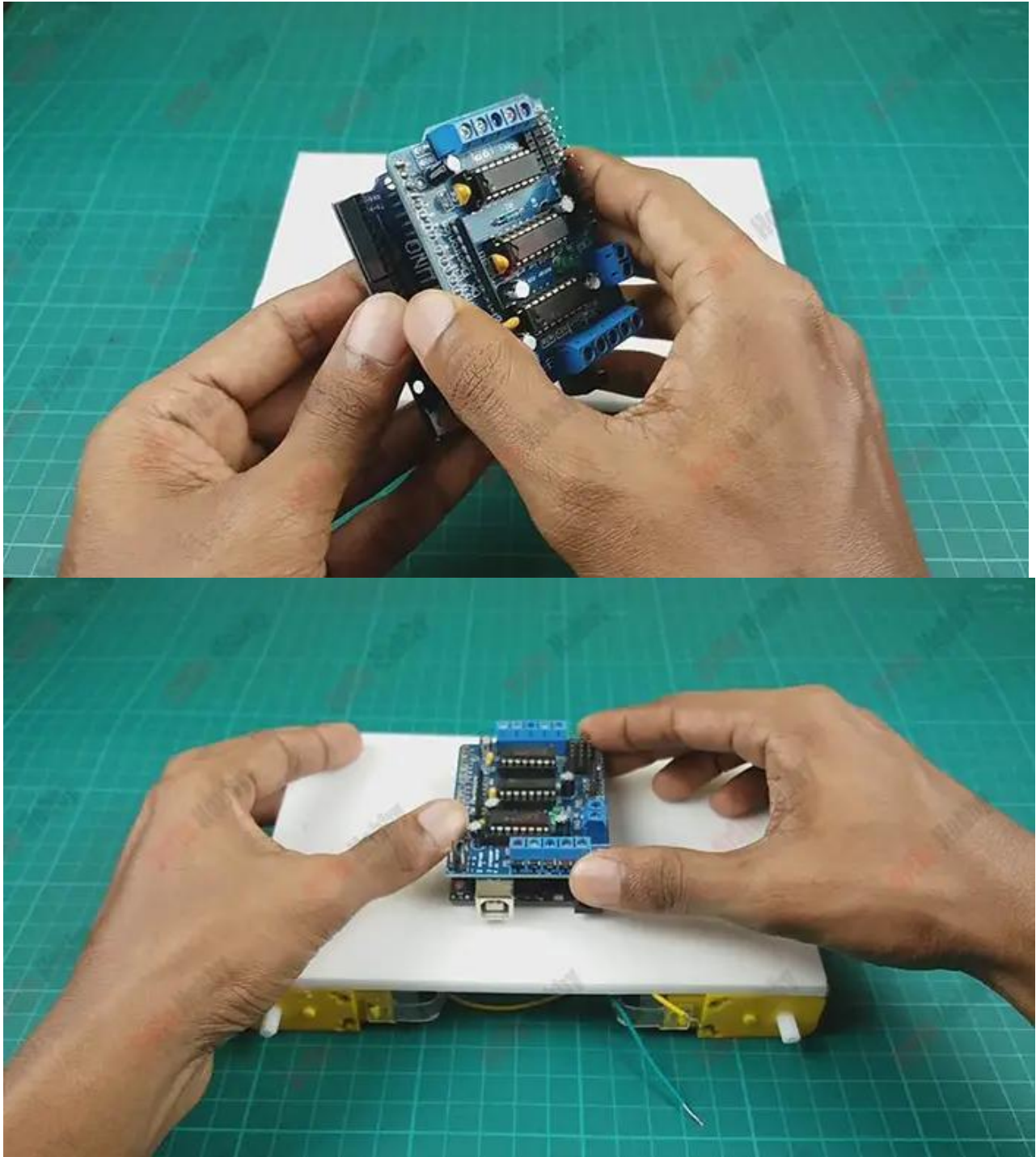
Step 3

Thirdly, glue the four gear motors to the foam board piece.



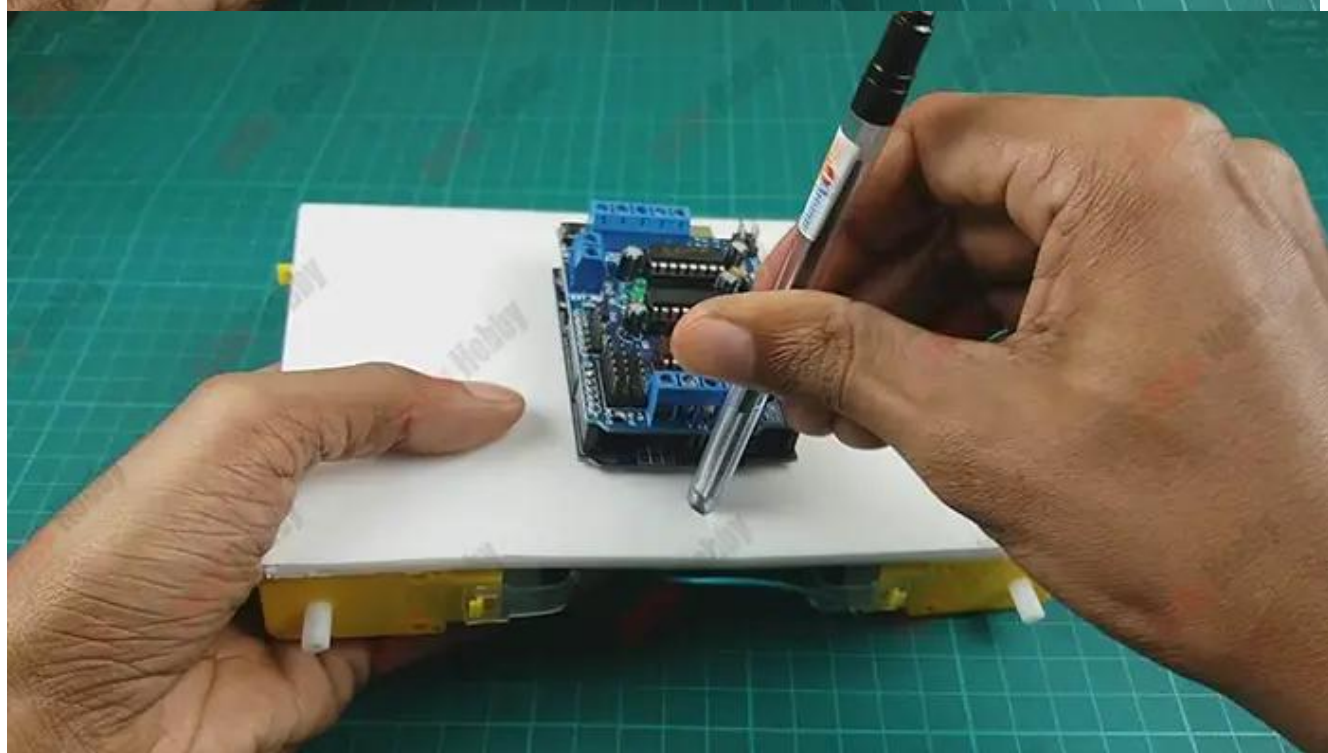
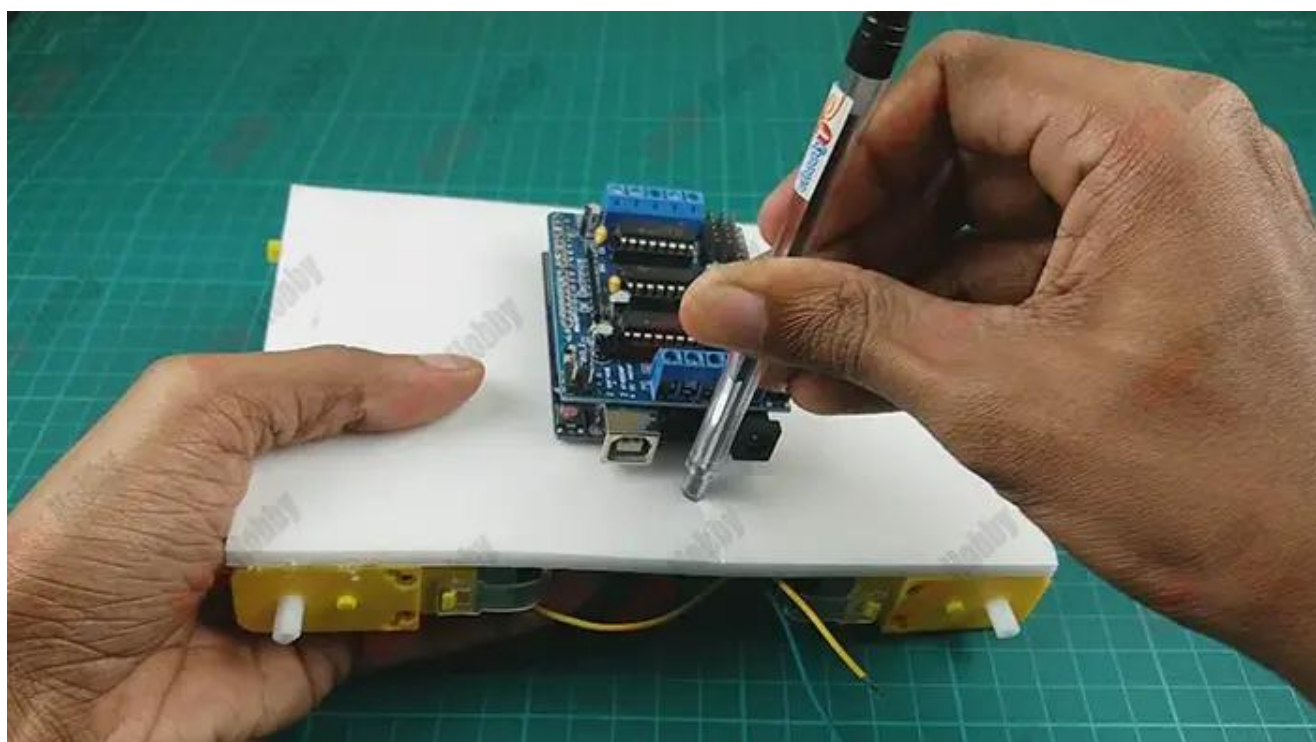
Step 4

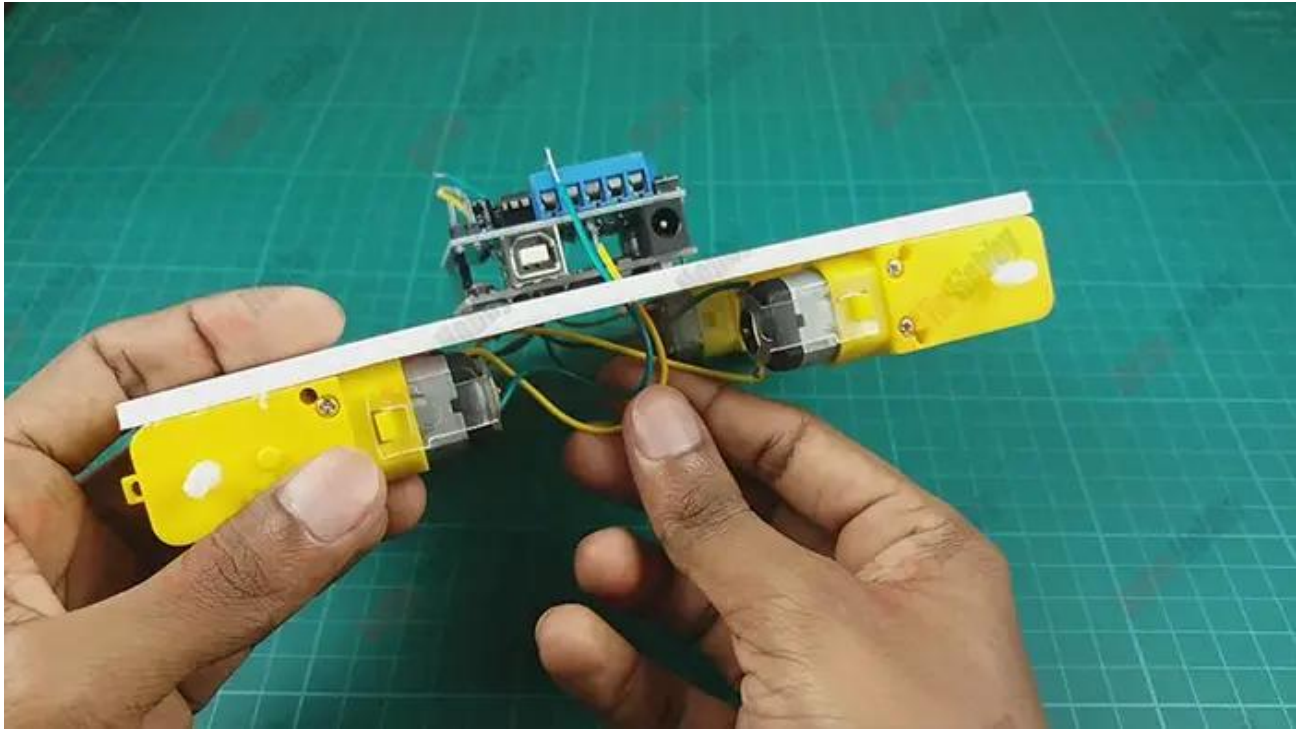
Then, attach the motor shield to the Arduino board and glue it to the robot chassis.



Step 5

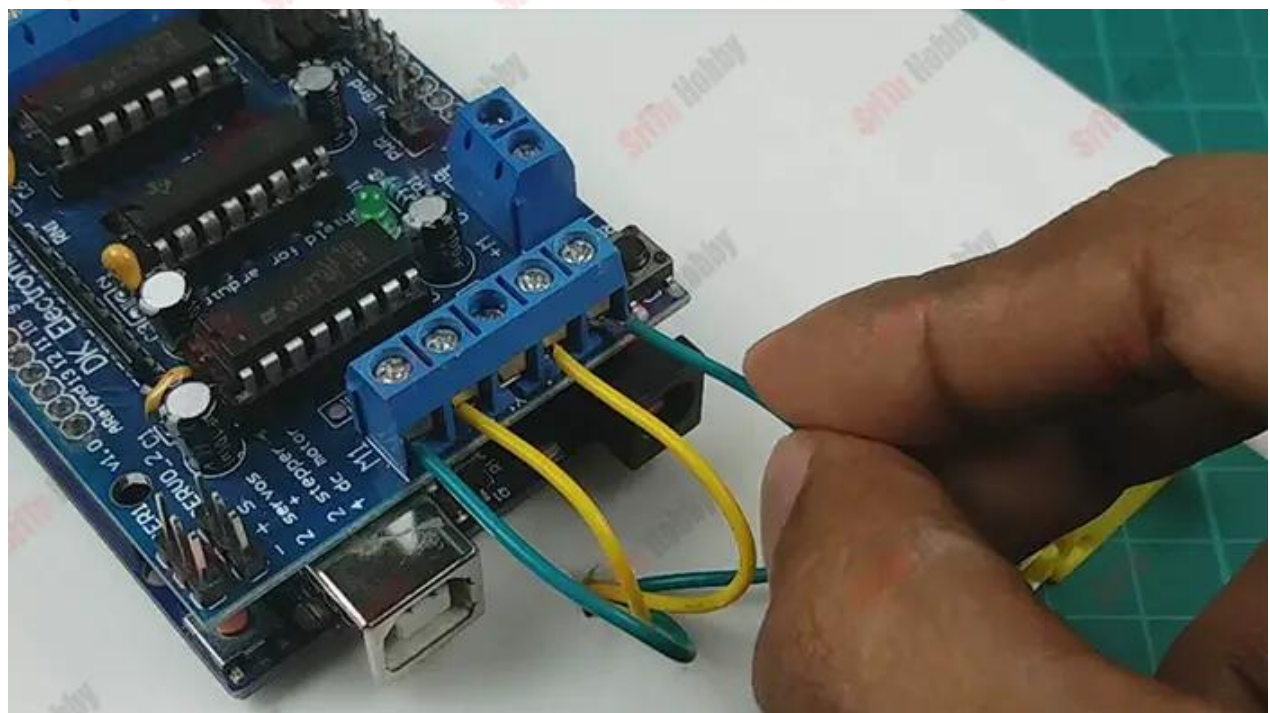
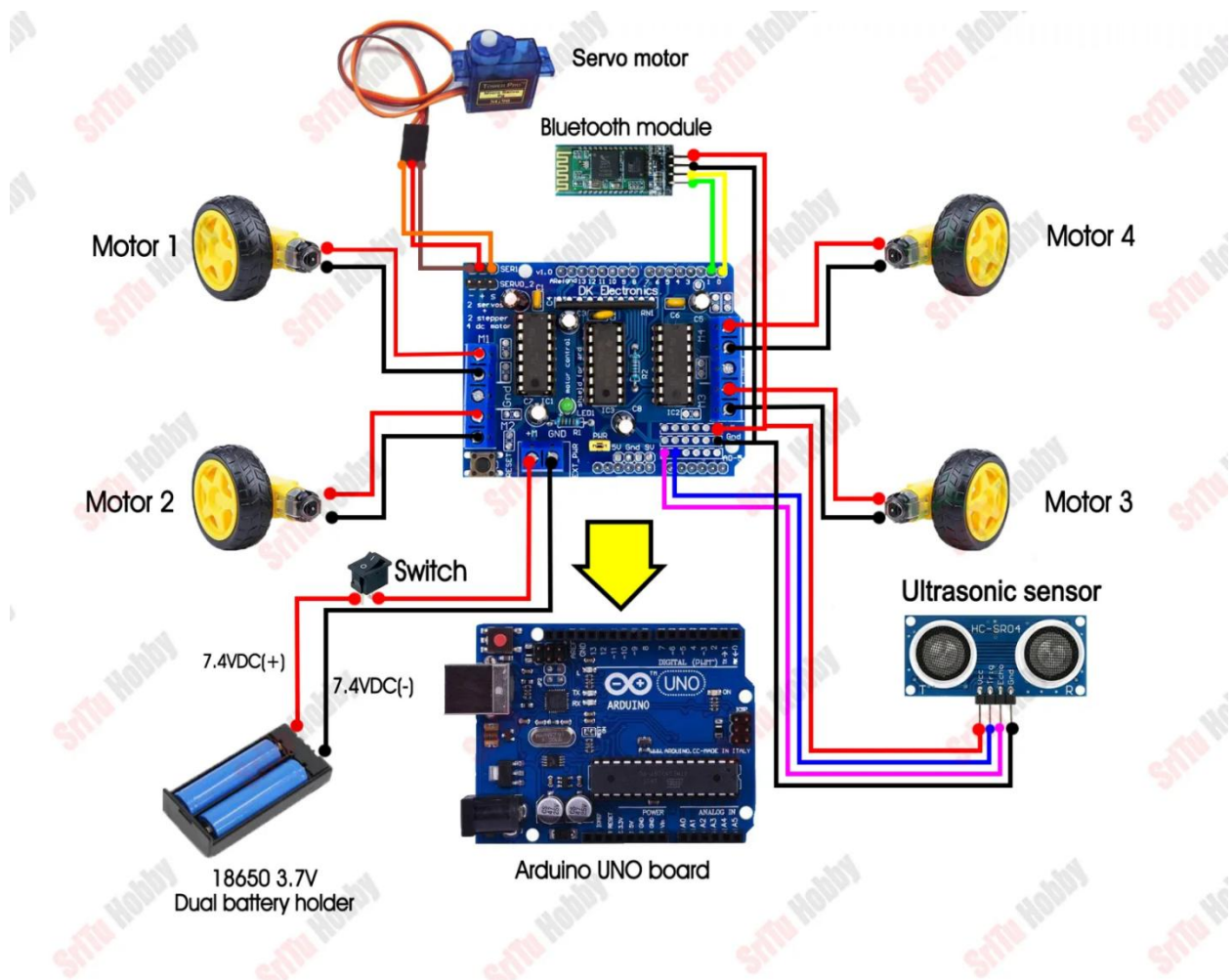
Next, dig two holes on either side of the Arduino board and insert the gear motor wire through these holes.





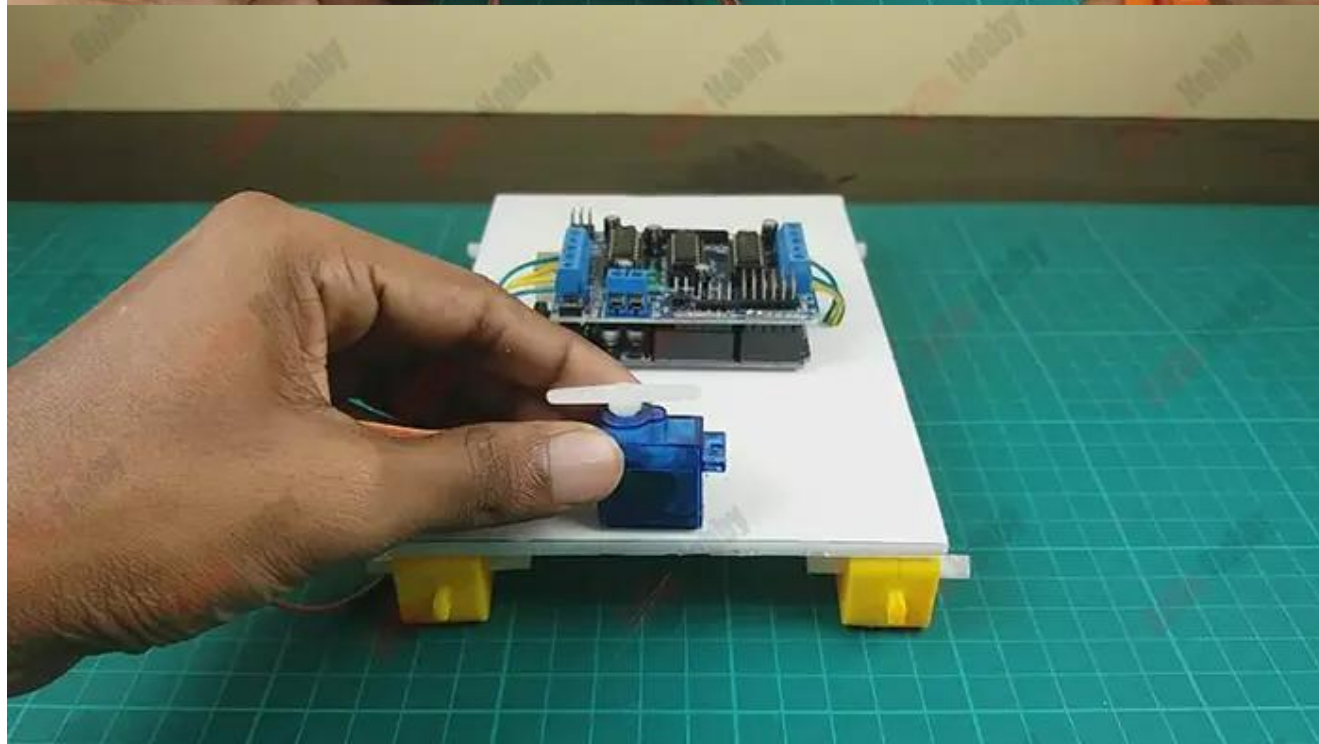
Step 6

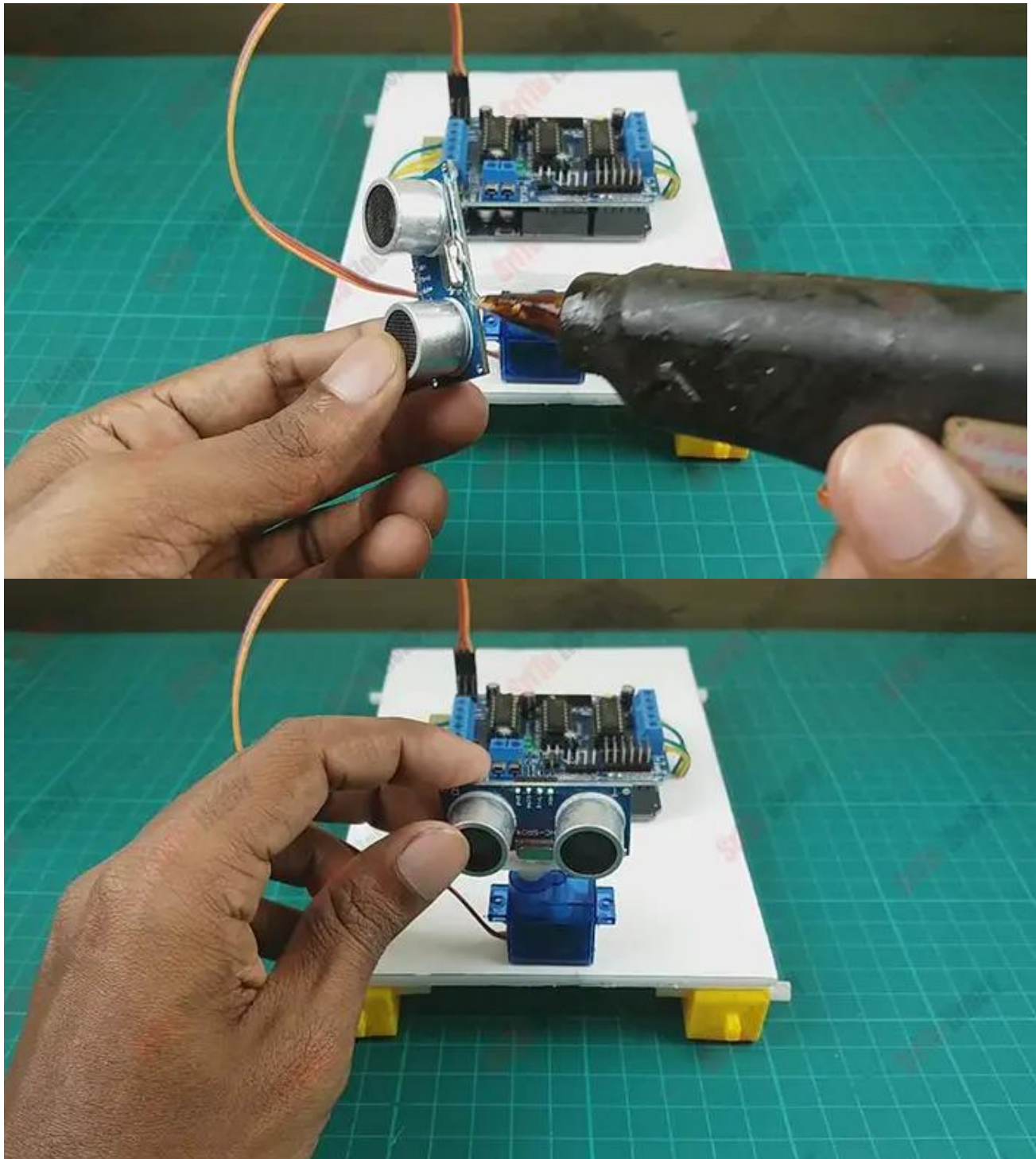
Then, connect the motors to the motor driver shield. To do this, use the circuit diagram below.



Step 7

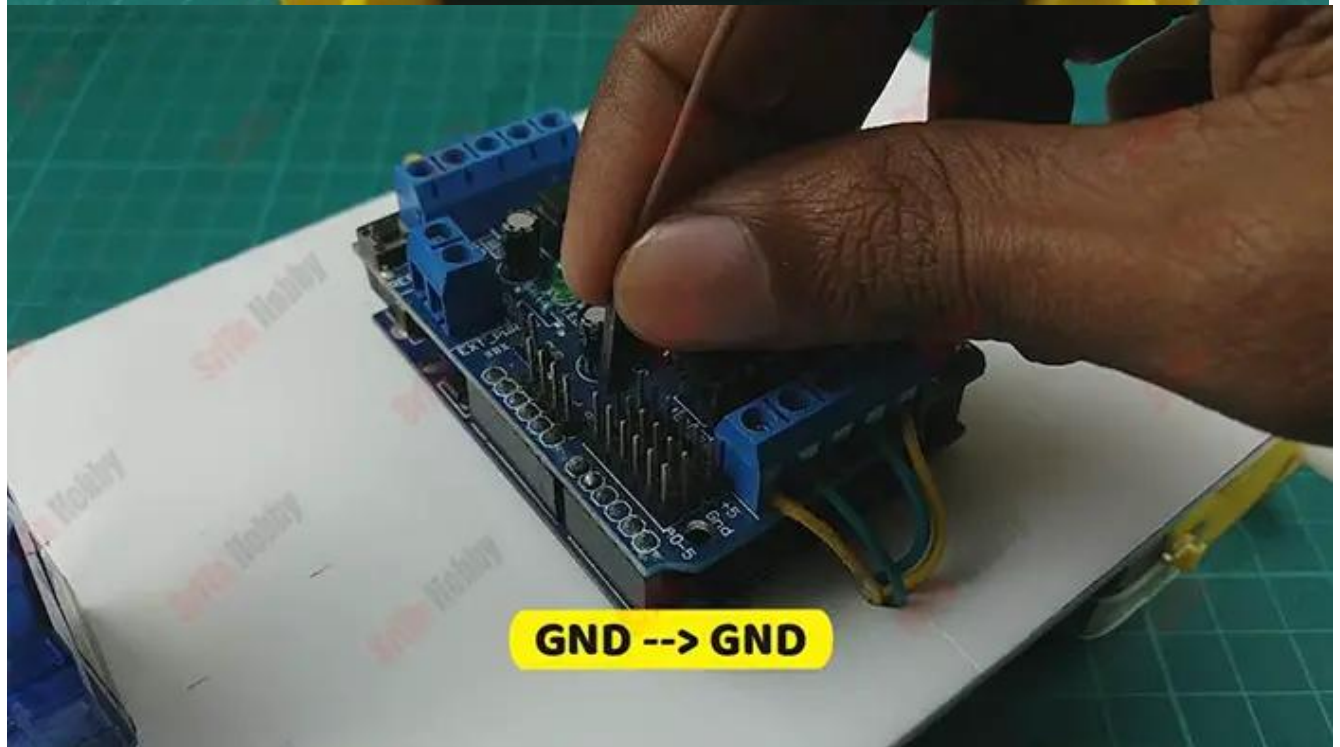
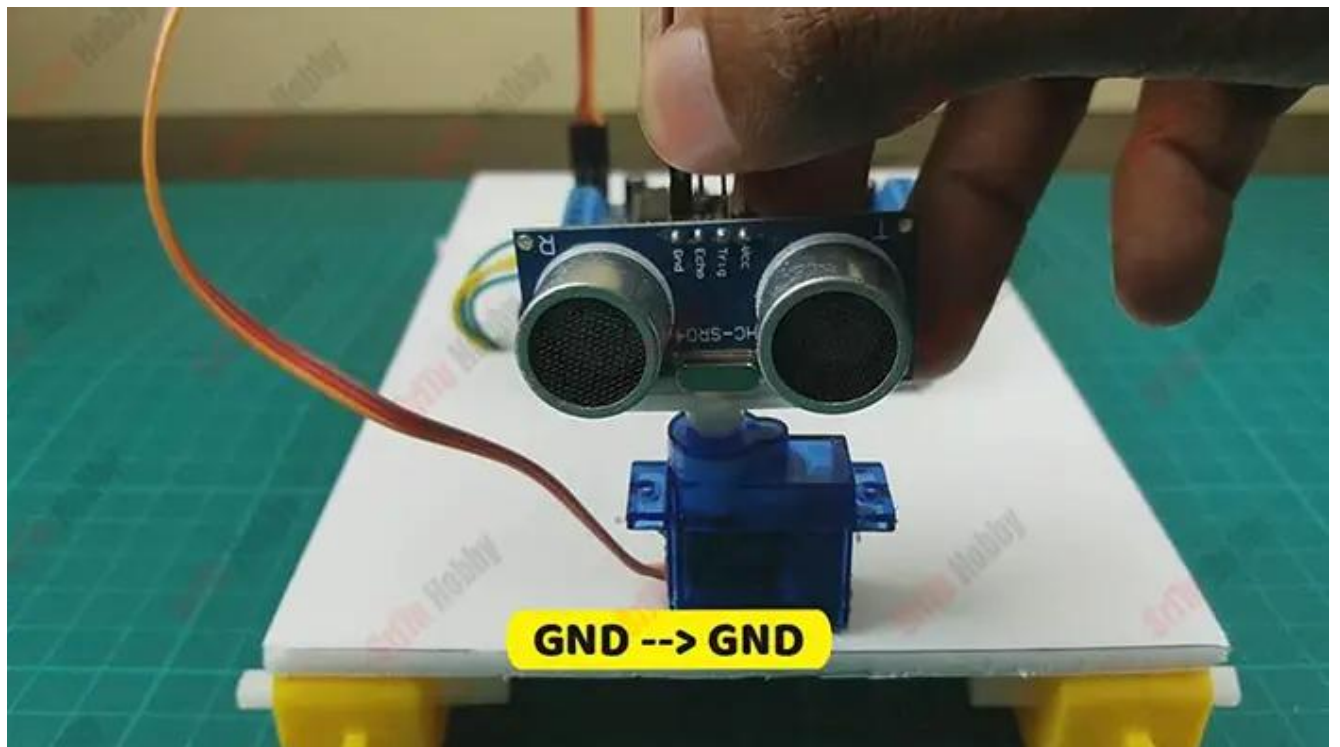
Afterward, attach the servo motor and ultrasonic sensor as follows.





Step 8

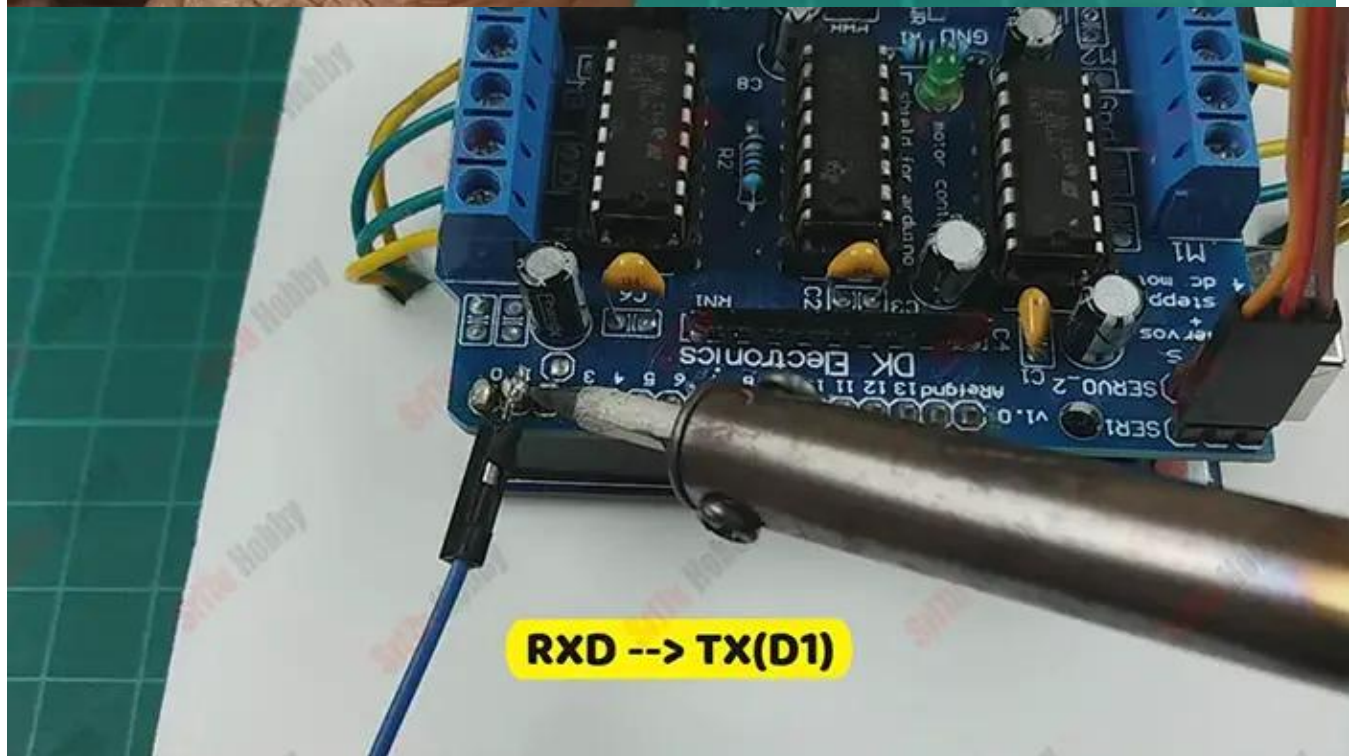
Next, connect the servo motor and the ultrasonic sensor using the circuit diagram above.

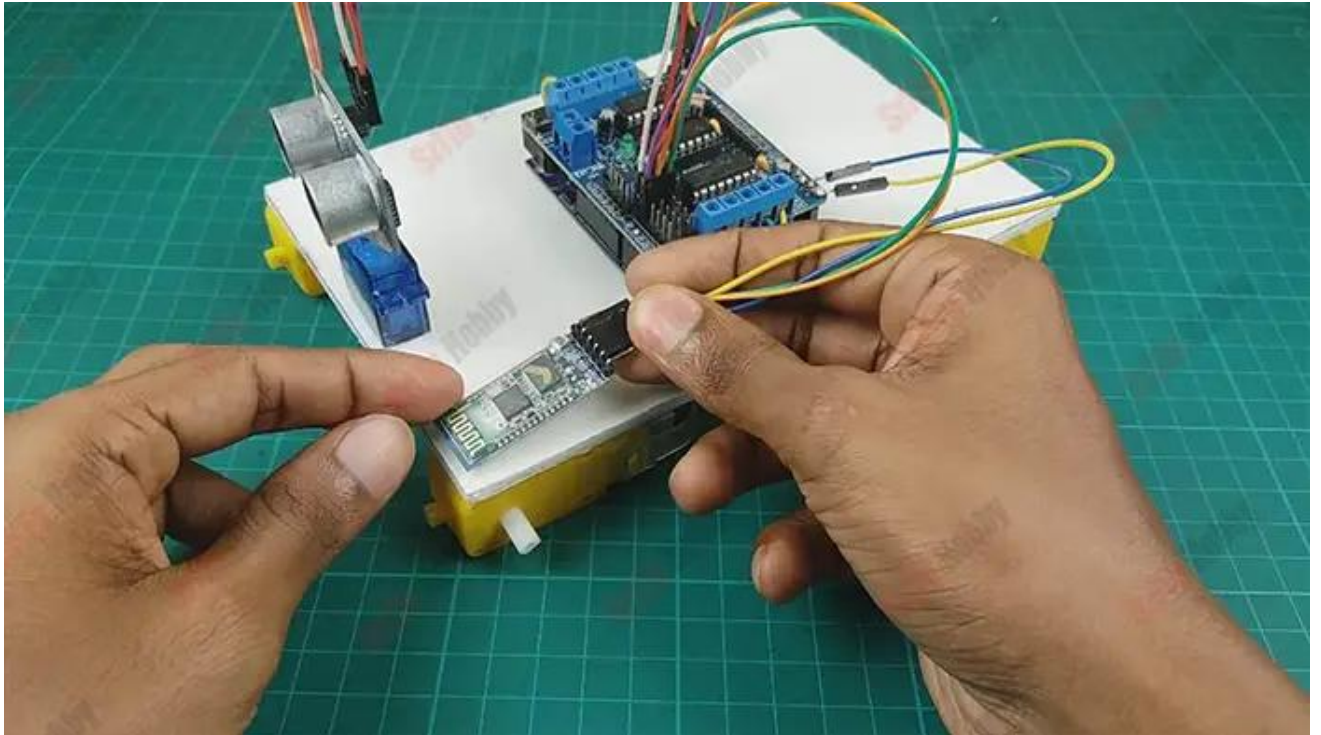




Step 9

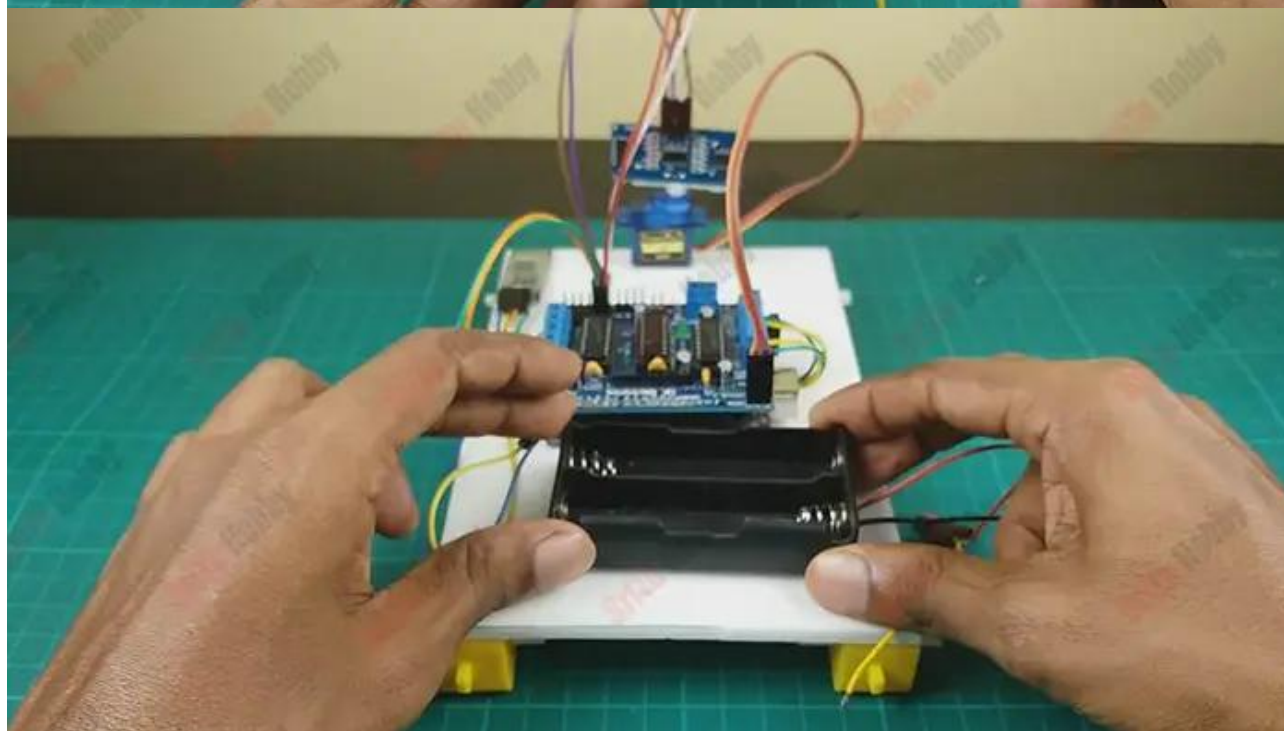
Then, connect the Bluetooth module to the motor driver shield and glue it to the robot chassis.

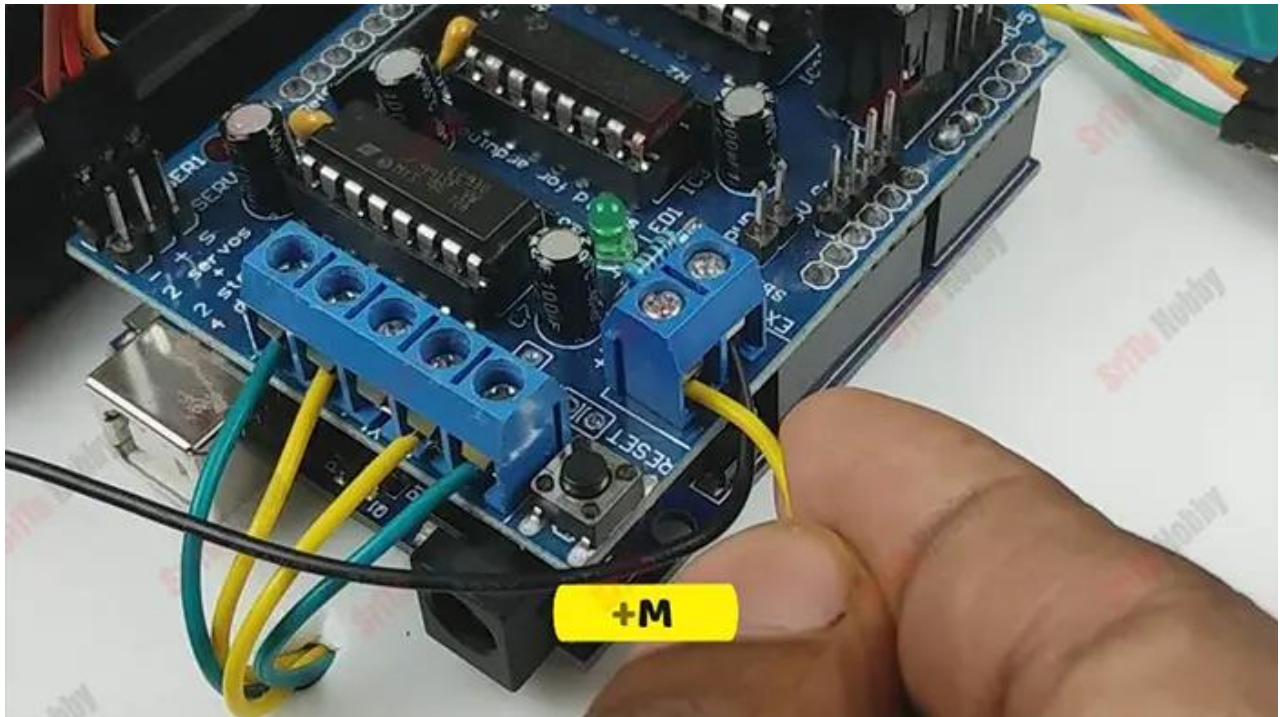




Step 10

After, glue the battery holder and connect it to the driver shield.





Step 11

Now, attach the robot wheels and put the batteries to the battery holder.



Step 12

So, let's create the program for this project. This program includes all three functions. We can run these separately. It is as follows.


```

/*obstacle avoiding, Bluetooth control, voice control robot car.

*/
#include <Servo.h>
#include <AFMotor.h>
#define Echo A0
#define Trig A1
#define motor 10
#define Speed 170
#define spoint 103
char value;
int distance;
int Left;
int Right;
int L = 0;
int R = 0;
int L1 = 0;
int R1 = 0;
Servo servo;
AF_DCMotor M1(1);
AF_DCMotor M2(2);
AF_DCMotor M3(3);
AF_DCMotor M4(4);
void setup() {
    Serial.begin(9600);
    pinMode(Trig, OUTPUT);
    pinMode(Echo, INPUT);
    servo.attach(motor);
    M1.setSpeed(Speed);
    M2.setSpeed(Speed);
    M3.setSpeed(Speed);
    M4.setSpeed(Speed);
}
void loop() {
    //Obstacle();
    //Bluetoothcontrol();
    //voicecontrol();
}
void Bluetoothcontrol() {
    if (Serial.available() > 0) {
        value = Serial.read();
        Serial.println(value);
    }
    if (value == 'F') {
        forward();
    } else if (value == 'B') {
        backward();
    } else if (value == 'L') {
        left();
    } else if (value == 'R') {
        right();
    } else if (value == 'S') {
        Stop();
    }
}
void Obstacle() {
    distance = ultrasonic();
    if (distance <= 12) {
        Stop();
    }
}

```

```

backward();
delay(100);
Stop();
L = leftsee();
servo.write(spoint);
delay(800);
R = rightsee();
servo.write(spoint);
if (L < R) {
    right();
    delay(500);
    Stop();
    delay(200);
} else if (L > R) {
    left();
    delay(500);
    Stop();
    delay(200);
}
} else {
    forward();
}
}
void voicecontrol() {
    if (Serial.available() > 0) {
        value = Serial.read();
        Serial.println(value);
        if (value == '^') {
            forward();
        } else if (value == '-') {
            backward();
        } else if (value == '<') {
            L = leftsee();
            servo.write(spoint);
            if (L >= 10 ) {
                left();
                delay(500);
                Stop();
            } else if (L < 10) {
                Stop();
            }
        } else if (value == '>') {
            R = rightsee();
            servo.write(spoint);
            if (R >= 10 ) {
                right();
                delay(500);
                Stop();
            } else if (R < 10) {
                Stop();
            }
        } else if (value == '*') {
            Stop();
        }
    }
}
}
// Ultrasonic sensor distance reading function
int ultrasonic() {
    digitalWrite(Trig, LOW);

```

```

    delayMicroseconds(4);
    digitalWrite(Trig, HIGH);
    delayMicroseconds(10);
    digitalWrite(Trig, LOW);
    long t = pulseIn(Echo, HIGH);
    long cm = t / 29 / 2; //time convert distance
    return cm;
}
void forward() {
    M1.run(FORWARD);
    M2.run(FORWARD);
    M3.run(FORWARD);
    M4.run(FORWARD);
}
void backward() {
    M1.run(BACKWARD);
    M2.run(BACKWARD);
    M3.run(BACKWARD);
    M4.run(BACKWARD);
}
void right() {
    M1.run(BACKWARD);
    M2.run(BACKWARD);
    M3.run(FORWARD);
    M4.run(FORWARD);
}
void left() {
    M1.run(FORWARD);
    M2.run(FORWARD);
    M3.run(BACKWARD);
    M4.run(BACKWARD);
}
void Stop() {
    M1.run(RELEASE);
    M2.run(RELEASE);
    M3.run(RELEASE);
    M4.run(RELEASE);
}
int rightsee() {
    servo.write(20);
    delay(800);
    Left = ultrasonic();
    return Left;
}
int leftsee() {
    servo.write(180);
    delay(800);
    Right = ultrasonic();
    return Right;
}

```


Code explanation

Firstly, libraries are included.

```
#include <Servo.h>
#include <AFMotor.h>
```

Secondly, ultrasonic sensor pins, servo motor pins, motor speed, and servo motor starting point are defined.

```
#define Echo A0
#define Trig A1
#define motor 10
#define Speed 170
#define spoint 103
```

Thirdly, some variables have been created to help the program.

```
char value;
int distance;
int Left;
int Right;
int L = 0;
int R = 0;
int L1 = 0;
int R1 = 0;
```

Then, objects are created for the Servo Library and the AFMotor Library.

```
Servo servo;
AF_DCMotor M1(1);
AF_DCMotor M2(2);
AF_DCMotor M3(3);
AF_DCMotor M4(4);
```

In the setup function, Ultrasonic pins are set to INPUT and OUTPUT. Also, the gear motor speeds have been included.

```
void setup() {
  Serial.begin(9600);
  pinMode(Trig, OUTPUT);
  pinMode(Echo, INPUT);
  servo.attach(motor);
  M1.setSpeed(Speed);
  M2.setSpeed(Speed);
  M3.setSpeed(Speed);
  M4.setSpeed(Speed);
}
```

In the loop function, the three main functions are included. we can run these functions one by one. These are described below.

```
void loop() {
  //Obstacle();
  //Bluetoothcontrol();
}
```

```

    //voicecontrol();
}

```

This function includes the Bluetooth control code. The code lines are described one by one in the code

```

void Bluetoothcontrol() {
//gets the serial communication values and puts them into the char variable.
    if (Serial.available() > 0) {
        value = Serial.read();
        Serial.println(value);
    }
//Next, these values are checked using the IF condition.
//Then, if the char value is 'F', the car moves forward.
    if (value == 'F') {
        forward();
//If the char value is "B", the car moves backward.
    } else if (value == 'B') {
        backward();
//If the char value is "L", the car moves left.
    } else if (value == 'L') {
        left();
//If the char value is "R", the car moves right.
    } else if (value == 'R') {
        right();
//If the char value is "S", the car is stopped.
    } else if (value == 'S') {
        Stop();
    }
}

```

This function includes the obstacle-avoiding code. The code lines are described one by one in the code.

```

void Obstacle() {

//gets the ultrasonic sensor reading and puts it into the variable.
    distance = ultrasonic();

//then, these values are checked using the IF condition.
//If the value is less than or equal to 12,
//the robot is stopped and the servo motor rotate left and right.
// Also, gets both side distance.
    if (distance <= 12) {
        Stop();
        backward();
        delay(100);
        Stop();
        L = leftsee();
        servo.write(spoint);
        delay(800);
        R = rightsee();
        servo.write(spoint);

//After, if the left side distance less than the right side distance. The robot
turns right.
        if (L < R) {
            right();

```

```

        delay(500);
        Stop();
        delay(200);

//After, if the left side distance more than the right side distance. The robot
turns left.
    } else if (L > R) {
        left();
        delay(500);
        Stop();
        delay(200);
    }

//Otherwise, the robot moves forward.
    } else {
        forward();
    }
}

```

This function includes the voice control code. The code lines are described one by one in the code.

```

void voicecontrol() {

//gets the serial communication values and puts them into the char variable.
    if (Serial.available() > 0) {
        value = Serial.read();
        Serial.println(value);

//If the char value is "^", the car moves forward.
        if (value == '^') {
            forward();

//If the char value is "-", the car moves backward.
        } else if (value == '-') {
            backward();

//If the char value is "<", the car moves left.
        } else if (value == '<') {
            L = leftsee();
            servo.write(spoint);
            if (L >= 10 ) {
                left();
                delay(500);
                Stop();
            } else if (L < 10) {
                Stop();
            }

//If the char value is ">", the car moves right.
        } else if (value == '>') {
            R = rightsee();
            servo.write(spoint);
            if (R >= 10 ) {
                right();
                delay(500);
                Stop();
            } else if (R < 10) {

```



```
        Stop();
    }

//If the char value is "*", the car is stopped.
    } else if (value == '*') {
        Stop();
    }
}
}
```

Step 13

Obstacle avoidance program

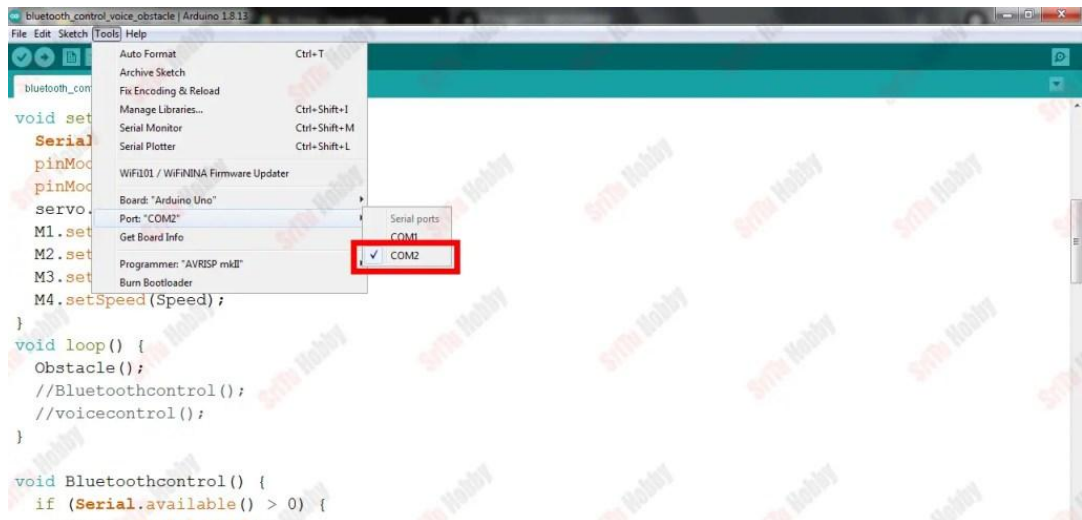
OK, now connect this robot car to the computer. Then, remove the two forward slashes in front of the “obstacle” function. Next, **remove the RX and TX jumper wires connected to the Bluetooth module.**



```
servo.attach(motor);  
M1.setSpeed(Speed);  
M2.setSpeed(Speed);  
M3.setSpeed(Speed);  
M4.setSpeed(Speed);  
}  
void loop() {  
  Obstacle();  
  //Bluetoothcontrol();  
  //voicecontrol();  
}  
  
void Bluetoothcontrol() {  
  if (Serial.available() > 0) {  
    value = Serial.read();  
    Serial.println(value);  
  }  
  if (value == 'F') {  
    forward();  
  } else if (value == 'B') {
```

Now, select board and port. After, upload this code to the robot and **reconnect the RX and TX jumper wires**.

-



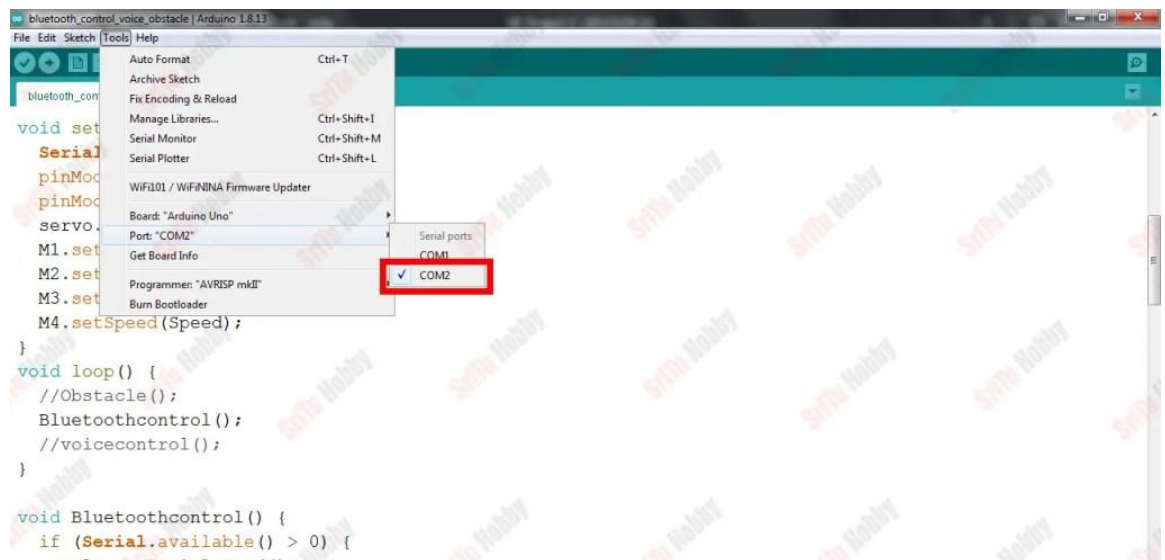
Now, power On this robot and enjoy it.

Bluetooth control program

OK, now connect this robot car to the computer. Then, remove the two forward slashes in front of the “Bluetooth control” function. Next, **remove the RX and TX jumper wires connected to the Bluetooth module.**



Now, select board and port. After, upload this code to the robot and **reconnect the RX and TX jumper wires**.





OK, now download and install the app below. Then, follow the steps below.



bluetooth control



4.0+ ★

4.5+ ★

New



Bluetooth finder: auto connect your device

Ad • Your Smart Technology • Tools

pair bluetooth android device

4.3 ★ 9.5 MB 10K+



Bluetooth Auto Connect – Devices Pair & Connect

Ad • bt autoconnect group • Tools

Controlling device statuses

3.5 ★ 16 MB 1M+



Arduino Bluetooth Control

broxcode • Tools

Installed



Arduino bluetooth controller

Giumig Apps • Tools

4.1 ★ 7.6 MB 500K+



Arduino Bluetooth RC Car

Andi.Co • Education

Installed



Bluetooth Control for Arduino

merahkemarun • Tools

4.1 ★ 1.6 MB 100K+



Bluetooth Controller HC-05

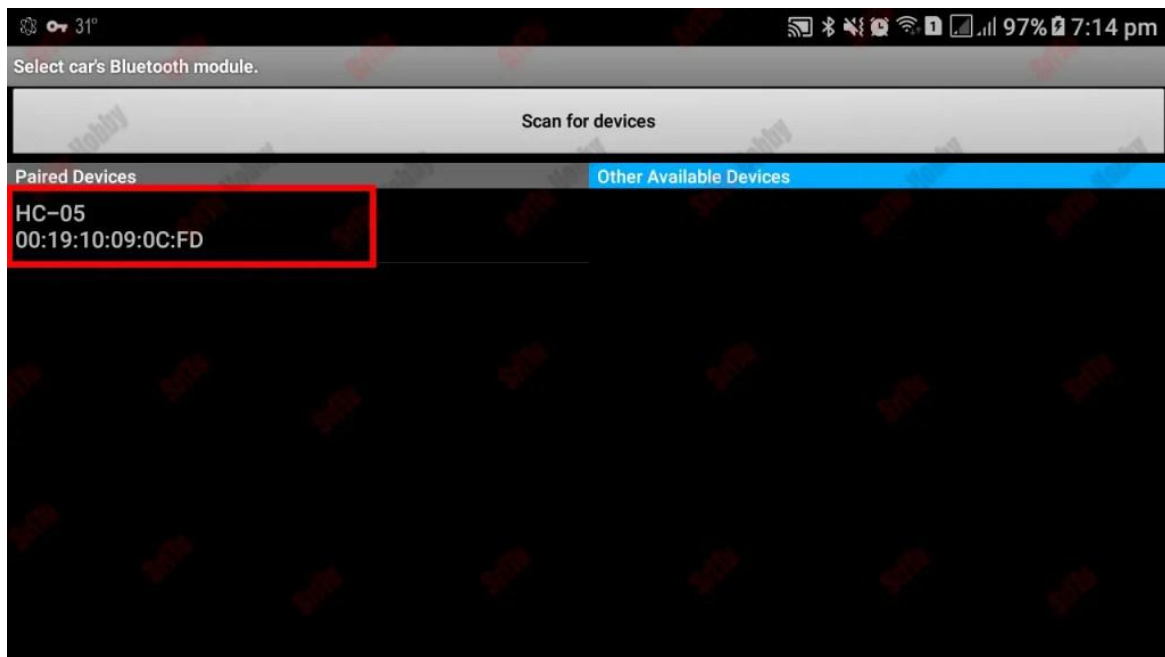
mightyIT • Communication

3.2 ★ 2.6 MB 100K+



Bluetooth Device Controller

After, run this application and click the Settings button. Then, click the “Connect to Car” button and select the name of the Bluetooth module. Now, you can see the green bulb in the corner.



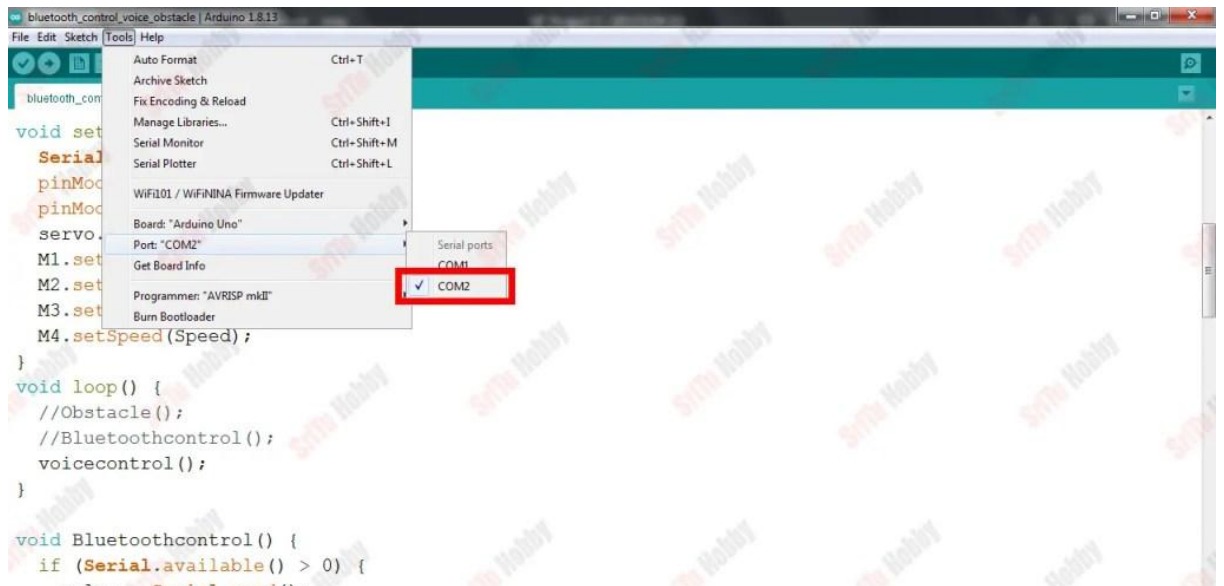
OK, now click the controller buttons and enjoy it.

Voice control program

OK, now connect this robot car to the computer. Then, remove the two forward slashes in front of the “voicecontrol” function. Next, **remove the RX and TX jumper wires connected to the Bluetooth module.**

Now, select board and port. After, upload this code to the robot and **reconnect the RX and TX jumper wires.**

-



OK, now download and install the app below. Then, follow the steps below.

After, run this application and click the setting button. Then, click the “voice commands configuration ” button and include the commands one by one. These are as follows.

Home



Arrow Keys



Terminal



Accelerometer



Buttons & Slider



Metrics



Voice Control

< Settings

When checked, will trigger the alarm if the received value is lower than the threshold.

☐

Shaker

Shake your device to control your arduino!

☐

Shaker sensitivity

Choose from the list

Shake data

Configure data to send while shake

BUTTONS AND SLIDER

Range

Set the range of your seekbar

Buttons sensitivity

Still send data while pressing

☐

Command buttons configuration

Configure data to send

VOCAL COMMANDS

Vocal commands configuration

Configure data to send

BE PART OF US

Join our facebook community and share with us your creativity !

Vocal command n° 1

Vocal command

Type the vocal command to pronounce

Data to send to arduino

A

Cancel

OK



En



1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p

@ # \$ % & ' ()
a s d f g h j k l

⬆ * " ' : ; ! ?
z x c v b n m ⬅ x

?123



English



Vocal command n°2

Vocal command

Type the vocal command to pronounce

Vocal command

BACK

Cancel

OK



En



1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p

@ # \$ % & ' ()
a s d f g h j k l

⬆ * " ' : ; ! ?
z x c v b n m ✖

😊
?123



,

English

.



Vocal command n°2

Vocal command

Type the vocal command to pronounce

Data to send to arduino

+

Cancel

OK



En



1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p

@ # \$ % & ' ()
a s d f g h j k l

^ * " , ; : ! ?
z x c v b n m

?123



,

English

.



31° ...

97% 7:15 pm

Vocal command n°3

Vocal command

Type the vocal command to pronounce

Vocal command

LEFT

Cancel

OK



En



1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p

@ # \$ % & ' + ()
a s d f g h j k l

^ * " ' : ; ! ?
z x c v b n m

?123



English



Vocal command n° 4

Vocal command

Type the vocal command to pronounce

Vocal command

RIGHT

Cancel

OK



En



1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p

@ # \$ % & ' ()
a s d f g h j k l

^ * " , ; : ! ?
_ z x c v b n m [] \ /

123



English



31° ...

97% 7:15 pm

Vocal command n°4

Vocal command

Type the vocal command to pronounce

Data to send to arduino

>

Cancel

OK



En



1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p

@ # \$ % & ' + ()
a s d f g h j k l

^ * " ' : ; ! ?
_ z x c v b n m

?123



English



Vocal command n° 5

Vocal command

Type the vocal command to pronounce

Vocal command

STOP

Cancel

OK



En



1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p

@ # \$ % & ' ()
a s d f g h j k l

↑ * " ' : ; ! ?
z x c v b n m ↵

?123



English



31° ...

96% 7:15 pm

Vocal command n° 5

Vocal command

Type the vocal command to pronounce

Data to send to arduino

*

Cancel

OK



En



1

2

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4

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y

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i

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&

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+

(

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j

k

l



z

x

c

v

b

n

m



?123



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English

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